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# Generating cheongsam custom pattern based on fuzzy set theory



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## Abstract

**Purpose** – The purpose of this paper is to quickly acquire a cheongsam pattern using the fit quantification method to meet individual fit requirement.

**Design/methodology/approach** – Based on the cheongsam pattern database including basic patterns and graded patterns, we defined the main control parts of the cheongsam pattern by analyzing the pattern modification. Combining human body shape characteristics, this paper utilized the fuzzy membership function to quantify the cheongsam fit, and defined the modified model of the cheongsam control part.

**Findings** – The fitness quantification method can provide suitable primary body characteristics for custom-pattern and helps to produce customized cheongsam quickly.

**Originality/value** – This paper proposed a method of generating customized cheongsam pattern based on fitness quantification by using fuzzy membership function. The method combined the industry pattern design experience and mathematic knowledge to generate the individual fit pattern rapidly. It can be applied in cheongsam customization.

**Keywords** Cheongsam, Industrial customization, Fuzzy membership function, Virtual fitting

**Paper type** Research paper

## Introduction

Due to the increasing emergence of individual fit demand, the industrial garment customization has become a hot spot in the academic and apparel industry. The customized pattern is designed for customers to choose their favorite products provided by the company, which makes the required garments according to the size of the customer's body (Peng, 2011). This kind of production method not only satisfies the user's body shape characteristics, but also meets the user's individualized needs. At the same time, it enables the enterprise to produce by sales without worrying about inventory, which caters to the market demand (Xu *et al.*, 2018).

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Cheongsam has higher fit requirements than other clothing. Most of the traditional cheongsam production is the single-piece version for customers (Ellen *et al.*, 2017; Liu and Xie, 2018; Xie and Yu, 2012). This production method has a long process chain and low efficiency, which cannot meet the growing demands of consumers and the rapid trends in the market. Industrial customization has the advantages of both body fit and high production efficiency, undoubtedly become a recommended manufacturing approach satisfies the people's demand.

The industrial customization methods normally can generate or modify the pattern based on template technology. There are some methods published or adapted in apparel industry. One is the neural network, used to establish the relation between human body size (Yu and Zhen, 2013) and garment, and provide detailed sizes for pattern making (Zhu *et al.*, 2017). Essentially, the network builds a relational model with parametric topological and dimensional constraints based on numerical and geometric constraints between the pattern parameter points, to implement pattern generation. Some researchers applied product family information the combination of style parts in the library to generate prototypes quickly (Li and Qi, 2008). However, the current research focus is mainly on the design of the style. As the key technology, the rapid assemble model of the part patterns is seldom introduced. There is other existing approach that producing individual pattern through grading operation based on the well-established pattern database (Han *et al.*, 2014). Basically, grading is an overall modification with a specific proportion. Customers who do not match the size need to edit the pattern individually, which cannot be modified immediately (Thomassey and Bruniaux, 2013).

Many apparel companies have accumulated a certain nimble of pattern data and grading experiences during many years of production. Based on the present pattern production, this paper proposed the method using the fuzzy set theory to solve the refinement of the pattern change value, and accomplished the rapid generation of the personalized cheongsam pattern based on the database of cheongsam pattern. This method solves the semantics ambiguity of the cheongsam pattern modification, reduces the subjective judgment and dependence of the pattern maker, and improves the fit of the cheongsam and the satisfaction of the customers. The specific customized technical route is shown in Figure 1.

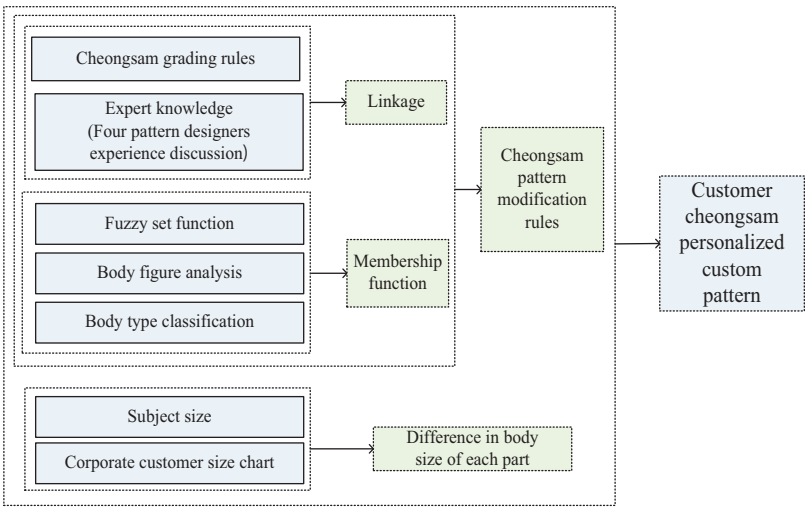


**Figure 1.**
Cheongsam custom technology workflow

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## Experimental subject and modified parts of the cheongsam pattern

*Experimental subject*

According to the Chinese standard GB/T 1335.2–2008, the human body shape is divided into four categories, as shown in Table 1. The three-dimensional body scanner is used to measure the subjects in the tights (Wang and Li, 2011), and the dimensional data of various parts of the human body can be obtained in more detail. The human body size table selected in the experiment of this paper is shown in Table 2. The first subject has a chest-to-waist difference of 19 cm, and the second subject has a chest-to-waist difference of 20 cm, both of which belonged to lean body.

*Experimental fabrics*

The experimental fabric for cheongsam clothing was selected from silk blended fabrics for testing. The blended fabrics were made from 95% tussah silk and 5% spandex. To quantify the characteristic properties of blended fabrics used for tests, yarn density, thickness, mass, stiffness and drape of fabrics were tested. These characteristic properties described the quantitative characterization of fabrics. Assessment items of fabrics were based on ISO standard methods. Density standards refer to ISO 7211–2:1984 (ISO 7211, 1984). Thickness standards refer to ISO 5084:1996 (ISO 5084, 1996). Mass standards refer to ISO 3801:1977 (ISO 3801, 1977). Stiffness ISO standards refer to 9073–7:1995 (ISO 9073, 1995). Drape standards refer to ISO 9073–9:2008 (ISO 9073, 2008). The specific values of the fabric properties are shown in Table 3.

*Modified parts of the cheongsam pattern*

In this paper, we choose the fitted cheongsam as the basic model for research. Firstly, we need to pick up the required samples in the pattern library. The components of the control part that affect the cheongsam fit are mainly the chest, waist, abdomen, buttocks and split ends (Lu, 2006; Hyun *et al.*, 2015).

| Body shape division | *Y* | *A* | *B* | *C* |
|---|---|---|---|---|
| Difference between chest and waist | 19–24 | 18–14 | 13–9 | 8–4 |
| Type of figure | Slim figure type | Normal figure | Busty figure | Fat figure |

**Note(s)**: Unit: cm

**Table 1.** Figure shape division

| Body parts | Height | Front chest | Back bust | Front waist | Back waist | Front abdominal | Back abdominal | Front hip | Back hip |
|---|---|---|---|---|---|---|---|---|---|
| 1# | 162 | 43 | 42 | 31 | 35 | 39.5 | 38 | 41 | 49 |
| 2# | 166 | 43.6 | 42.4 | 32.5 | 36.5 | 40 | 38 | 41.5 | 48.5 |

**Note(s)**: Unit: cm

**Table 2.** Test human body size

| Experimental items | Warp density/ Yarns/ 10cm | Weft density/ Yarns/ 10cm | Thickness mm | Mass g/m$^2$ | Stiffness-bending rigidity cN cm | Drape coefficient % |
|---|---|---|---|---|---|---|
| Blended fabric | 525 | 432 | 0.14 | 98 | 0.32 | 16 |

**Table 3.** Experimental fabric properties

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The control part of the cheongsam is closely related to its unique pattern. Compared with the industrial pattern of other types of clothing, the cheongsam customization needs additional parts such as abdominal circumference, splitting, and collar height, in which the circumference is subdivided into front and rear circumferences. The abdominal circumference is mainly for some customers with belly and convexity. After definition, it can be easily and quickly modified. The split ends are divided into a fork top width and a fork top height. The fork top width represents the circumference of the bodice at the apex of the split fork, and the fork top height represents the length of the split fork; in terms of circumference, for example, the chest is divided into a front chest and rear chest. When the customer's chest is larger than the standard level, two situations should be considered. One case is that the customer's chest cavity is wider than standard body's, then both the front and back chest should be modified. It may also result from plump chest, then the front chest should be modified.

Through discussions with four board masters with more than 20 years of work experience, the structural design theory is combined with the actual production of the factory, and the modified parts of the cheongsam pattern are defined from the three aspects of circumference, length and details (shown in Table 4). The control points covered by the modified part are subsequently encoded to quickly update the cheongsam pattern.

## Modification of cheongsam model based on fuzzy set theory

### *Modified part of the cheongsam pattern*

In this paper, the experimental sample is a sleeveless cheongsam. Through a deep understanding of the cheongsam pattern of the knowledge and discussions with the pattern of the master, we summed up the modified chest, waist, abdominal circumference and hip circumference of the linkage and control points, as shown in Figure 2.

According to the principle of rapid generation of pattern, the fewer the number of control points of the pattern is adjusted under the premise of ensuring the quality of the modification, the higher the efficiency of pattern generation. The chest circumference and waist circumference are the core parameters in the modification process due to the number distribution of linkage parts and control points, followed by the abdominal circumference and the hip circumference.

According to the human body sizes of the two experimental subjects 1# and 2#, the enterprise customer size and the cheongsam model size table (as shown in Table 5 below), the sample code numbers of this experiment were 6 and 8 yards respectively. The sizes of the posterior chest circumference, the anterior abdominal circumference, the posterior abdominal circumference and the posterior hip circumference corresponding to the version are the closest to the human body data, with less template modification and fewer control points.

### *Determination of modified value based on fuzzy set theory*

The test subjects 1# and 2# size, the corporate cheongsam body size and cheongsam sample size were compared for each part of the data. Note from Table 6 that the specific value of the subject's 1# part is subtracted from the specific value of the 6-yard human body size. The

**Table 4.** Cheongsam modification

| | |
|---|---|
| Circumference | Front bust girth, rear bust girth, breast spacing, front waist circumference, back waist circumference, front abdominal circumference, back abdominal circumference, front hip circumference, back hip circumference, front chest width, back width, fork width, shoulder width, collar width, cuffs Hem |
| Length | length, fork height, collar height, sleeve length |
| Details | front waist dart length (wide), back waist dart length (wide), front chest dart length (wide), underarm dart length (wide) |

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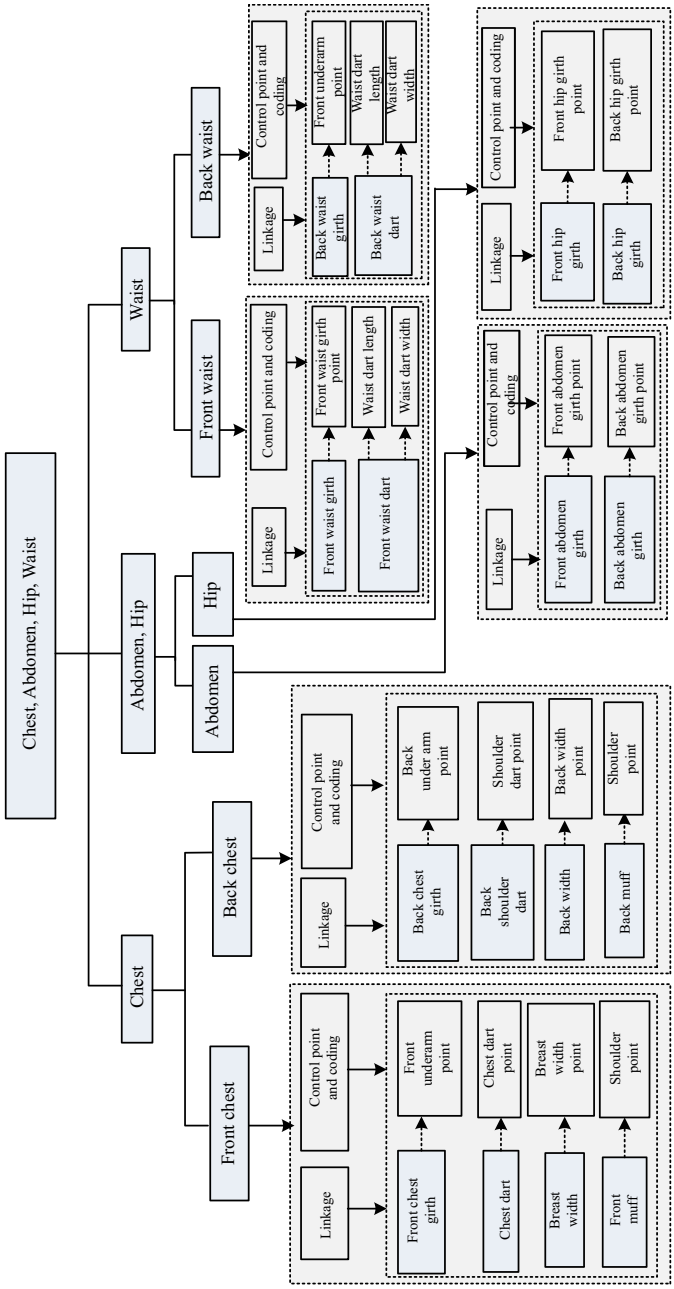


**Figure 2.**
Linkage and Control points

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**Table 5.**
Customer size and cheongsam pattern size of enterprise

| | Size/ Part | Height/Dress length | Front chest | Back bust | Front waist | Back waist | Front abdominal | Back abdominal | Front hip | Back hip |
|---|---|---|---|---|---|---|---|---|---|---|
| Customer Size | 6-yard | 160 | 43 | 41 | 33 | 35 | 38 | 36 | 43 | 47 |
| | 8-yard | 165 | 45 | 43 | 35 | 37 | 40 | 38 | 45 | 49 |
| Cheongsam pattern | 6-yard | 125 | 45 | 44 | 38 | 37 | 43 | 39 | 45 | 51 |
| | 8-yard | 130 | 47 | 46 | 40 | 39 | 45 | 41 | 47 | 53 |

**Note(s)**: Unit: cm

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result is: posterior chest circumference +1 cm, front waist circumference −2 cm, anterior abdominal circumference +1.5 cm, posterior abdominal circumference +2.5 Cm, front hip circumference −2 cm, rear hip circumference +2 cm. Next, note from Table 7 that the specific value of the second part of the subject is subtracted from the specific value of the body size of the 8 yards. The specific differences of the modified parts are: bust front −1.4 cm, front waist −4.5 cm, rear waist −1.5 cm, front hip −3.5 cm. The body characteristics of the subjects are flat chest, thin waist, slightly convex abdomen and rounded buttocks, which are inconsistent with the human body size in the human body size table. The corresponding size cannot be found in the standard size chart, and cannot be directly modified according to the difference. Therefore, this paper establishes the combined membership function based on the difference of body shape characteristics, and then the proportion of each part in the pattern can be changed.

Definition: Let $M$ be a map of set $D$ to [0, 1], $M{:}D \rightarrow [0, 1]$, $d \rightarrow M(d)$, then $D$ is the fuzzy set on $M$, and $M(d)$ is called the membership function of fuzzy Set $M$. The lean body type is defined as a fuzzy set $M$. When the chest and waist difference in the body type is getting larger and larger, the body type belongs to the set and becomes more and more unsatisfactory. By varying the degree of membership function values, we can quantitatively evaluate changes in body shape characteristics. The body shape corresponding to the cheongsam model is mainly determined by the circumference of the human body. The difference in body shape is mainly determined by the difference of the circumference between the chest circumference, the waist circumference, the abdominal circumference and the hip circumference. The change in body shape is proportional to the difference in circumference between the chest and the waist (Ni *et al.*, 2014; Young and Yun, 2010). Regardless of the lean body or the fat body, after the difference between the chest and the waist reaches a certain value, the body state characteristics will show a trend of almost no change in this body type range. By analyzing the relationship between the body side angle and the thoracolumbar

| Part/Category | 1# | Corporate cheongsam body 6 yards size | 6-yard cheongsam pattern size |
|---|---|---|---|
| Front chest | 43 | 43 | 45 |
| Back chest | 42 | 41 | 44 |
| Front waist | 31 | 33 | 38 |
| Back waist | 35 | 35 | 37 |
| Front abdominal | 39.5 | 38 | 43 |
| Back abdominal | 38.5 | 36 | 39 |
| Front hip | 41 | 43 | 45 |
| Back hip | 49 | 47 | 51 |

**Note(s)**: Unit: cm

**Table 6.** 1# size comparison table

| Part/Category | 2# | Corporate cheongsam body 8 yards size | 8-yard cheongsam pattern size |
|---|---|---|---|
| Front chest | 43.6 | 45 | 47 |
| Back bust | 42.4 | 43 | 46 |
| Front waist | 32.5 | 35 | 40 |
| Back waist | 36.5 | 37 | 39 |
| Front abdominal | 40 | 40 | 45 |
| Back abdominal | 38 | 38 | 41 |
| Front hip | 41.5 | 45 | 47 |
| Back hip | 48.5 | 49 | 53 |

**Note(s)**: Unit: cm

**Table 7.** 2# size comparison table

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difference, the change in body shape, the change in chest and waist difference can be approximated as a logarithmic function change:

$$p = \log_a^d + \mathrm{b} \tag{1}$$

Where $p$ represents the membership value and d represents the chest-waist difference. According to the body type division Table 1, the maximum value of the conventional body chest and waist difference d is 18 cm, the type corresponding to the body type is not modified, the membership value $p$ of the fuzzy set $M$ is 0; the maximum value of the lean body d is 24 cm, the time is already the upper limit of the chest-waist difference of the body type, and the membership value $p$ of the pair of fuzzy sets $M$ can be set to 1. By bringing (24,1), (18,0) into the formula (*) and calculating the membership function coefficients $a = 4/3$, $b = -\log_{4/3}^{18}$, the membership function of the body shape change set in this paper is established:

$$M(d) = \log_{4/3}^{d} - \log_{4/3}^{18} \quad (d \geq 18) \tag{2}$$

The chest-waist difference values of the experimental subjects 1# and 2# were brought into the equation (#), and the membership value $M$ ($d$) was calculated to be 0.19 and 0.37, respectively.

According to the definition of the cheongsam style and the linkage part, the number of linkage points of the figure sample was determined to be 47, corresponding to the linkage point to form the corresponding circumference, and the linkage point is encoded by numbers, as shown in Plate 1.

The number of linkage points of the posterior chest circumference, front waist circumference, anterior abdominal circumference, posterior abdominal circumference, front hip circumference and rear hip circumference of the experimental subject 1# are 6, 6, 2, 2, 2, 2, respectively. The number of linkage points is 47, corresponding to the linkage points to form

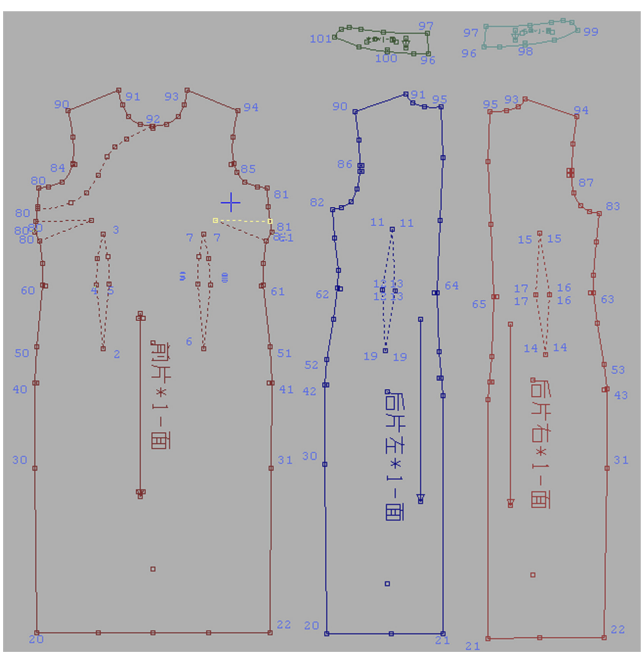

**Plate 1.**
Linkage point encoding

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the corresponding circumference; the chest circumference, waist circumference, abdominal circumference and hip circumference of the experimental subjects 1# are 85, 66, 78, 90 cm respectively; the circumferences of the posterior chest circumference, the front waist circumference, the anterior abdominal circumference, the posterior abdominal circumference, the front hip circumference and the rear hip circumference of the test subjects are 42, 31, 39.5, 38.5, 41, and 49 cm, respectively. The total area ratio of the parts is 0.49, 0.47, 0.51, 0.49, 0.46, and 0.54, respectively. The modification amount corresponding to each part in the 6-code pattern is shown in Table 8. The expectation of the experimental subject 1# on the cheongsam pattern modification is a fit type, and the modified results of the pattern meet the requirements of the subject.

The number of linkage points of the front chest circumference, front waist circumference, back waist circumference and front hip circumference of the second experimental subjects is 6, 10, 12, 2, respectively, and the number of linkage points of the body sample is 47 and corresponding linkage points are connected to form a corresponding circumference; the chest circumference, waist circumference, abdominal circumference and hip circumference of the experimental subjects are 86, 69, 78 and 90 cm respectively. The circumferences of the front chest circumference, front waist circumference, back waist circumference and front hip circumference of the test subjects are 43.6, 32.5, 36.5, 41.5 cm respectively, the total circumference ratio of each part is 0.51, 0.47, 0.53, 0.46. The amount of modification corresponding to each part in the 8-yard pattern is shown in Table 7. The experimental subject 2#'s expectation for the modification of the cheongsam pattern belongs to the body type, and the pattern can be modified to meet the requirements of the subject.

The input is modified value of each part into the defined linkage part. The original pattern and modified pattern of the test subject 1# are shown in Plate 2. The original pattern of the test subject 2# and the modified pattern are shown in Plate 3. The original pattern is in on the left, the modified pattern is on the right.

## Experimental result

In order to evaluate the modified method of the cheongsam custom template proposed in this paper, the cheongsam costumes of the modified pattern were made, and the fit evaluation of the real people's dress was carried out. At the same time, the virtual fitting software was imported and compared (Xu and Ding, 2007).

### *Physical dressing evaluation*

The cheongsam garment was produced by outputting 6-code pattern and using micro-elastic material blended with silk and spandex. The comparison chart of the testers before and after the modification of the pattern is shown in Plate 4. From the overall view of Plate 4, the modified pattern fit is significantly better than the original pattern.

From the front 6-yard pattern try-on picture (Plate 4a, it can be observed that the gap between the waist and the hip circumference of the cheongsam garment made by the original model is small, and the body angle of the wearer is small. From the side (Plate 4b, the cheongsam after the 6-yard pattern is put on a straight tube, and the amount of relaxation is higher in the waist and front hip. By adjusting the amount of relaxation of the front waist circumference, the front hip circumference and the back hip circumference, the modified cheongsam have a significant improvement in the fit of the waist and hip than the original pattern, and the cheongsam has a higher degree of fit. From the back (Plate 4c, after trying on the cheongsam of the 6-yard pattern, the human body has a certain fold on the back and back abdomen, and the angle of the side angle of the waist is small. By increasing the amount of relaxation of the abdominal circumference and the back chest circumference, the modified

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**Table 8.** Modification formula and modification amount of each part of the subject

| Subject 1# | | | Subject 2# | | |
|---|---|---|---|---|---|
| Linkage | Body size | Modification | Linkage | Body size | Modification |
| Back bust | 42 | $(+1.0)+1.0*(-0.19*0.49)\approx+0.90$ | Front bust | 43.6 | $(-1.4)+1.4*(-0.37*0.51)\approx-166$ |
| Front waist | 34 | $(-2.0)+2.0*(-0.19*0.52)\approx-2.20$ | Front waist | 32.5 | $(-2.5)+2.5*(-0.37*0.47)\approx-2.93$ |
| Anterior abdominal | 39.5 | $(+1.5)+1.5*(-0.19*0.51)\approx+1.36$ | Back waist | 36.5 | $(-0.5)+0.5*(-0.37*0.53)\approx-0.60$cm |
| Posterior abdominal | 38.5 | $(+2.5)+2.5*(-0.19*0.49)\approx+1.92$ | Front hip | 41.5 | $(-3.5)+3.5*(-0.37*0.51)\approx$-4.16 |
| Front hip | 41 | $(-2.0)+2.0*(-0.19*0.46)\approx-2.17$ | | | |
| Back hip | 49 | $(+2.0)+2.0*(-0.19*0.54)\approx+1.79$ | | | |

**Note(s)**: Unit: cm

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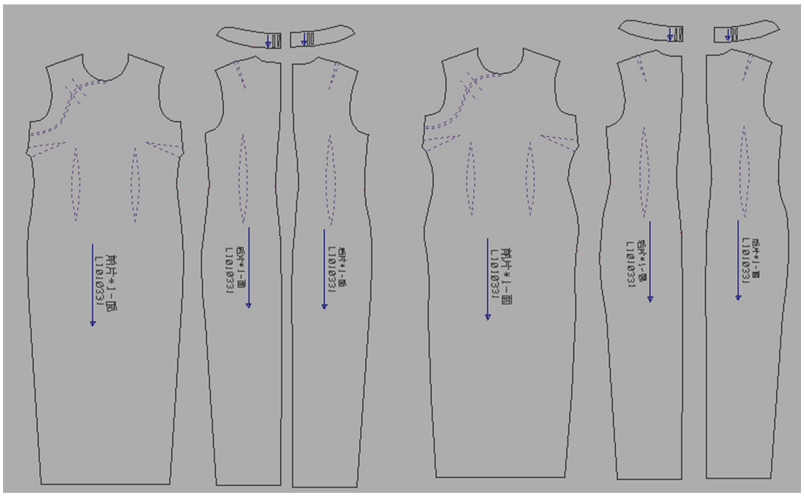


**Plate 2.**
The original pattern and the modified pattern

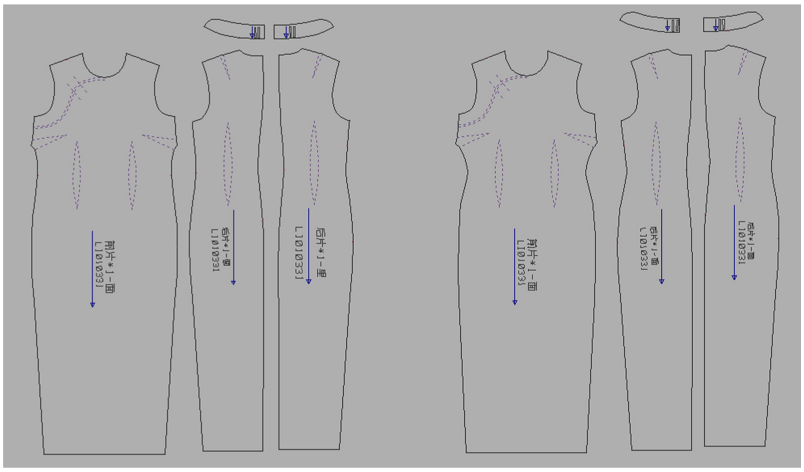


**Plate 3.**
The original pattern and the modified pattern

cheongsam of the modified pattern is more suitable for the back, and the angle of the side angle of the waist body is larger, and the whole back is relatively flat. From the modified wearing pattern of the modified model in Plate 4, cheongsam is more suitable in the chest, waist, abdomen and buttocks, and the overall effect of the garment is satisfactory.

The cheongsam garments were produced by outputting 8 yards of paper samples with the same fabric. The comparison of the testers before and after the modification of the model is shown in Plate 5. From Plate 5, the modified pattern fit is significantly better than the original. From the front of Plate 5a, it can be observed that the cheongsam garment has obvious wrinkles at the front breast area, the lower chest circumference and the front waist circumference with excess relaxation amount. The front chest and the front waist are more conformable by reducing the amount of relaxation of the front chest circumference, the front waist circumference, and the back circumference. From the side of Plate 5b, the cheongsam after the 8-yard pattern has been put on a straight tube shape, and the amount of relaxation in

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**Plate 4.**
Comparison of the cheongsam 6 yard pattern and modified pattern of the model

**Note(s):** (a) Try on the front (b) Try on the side (c) Try on the back

**Plate 5.**
Comparison of the cheongsam 8 yard pattern and modified pattern of the model

**Note(s):** (a) Try on the front (b) Try on the side (c) Try on the back

the chest, waist and buttocks is more. By adjusting the amount of relaxation of the front chest circumference, the front waist circumference and the front hip circumference, the modification is made. The cheongsam has a significant improvement in the fit of the chest, waist and hips compared to the original pattern. From the back of Plate 5c, after trying on the cheongsam of the 8-yard pattern, the human body has a certain fold on the back and there is excess relaxation, and the angle of the side angle of the waist is small. By adjusting the amount of relaxation of the back bust and the back waist circumference, the modified cheongsam of the modified model is relatively flat on the back, and the side angle of the waist body is larger, and the back is more integrated. From the try-on pattern of the modified pattern in Plate 5, the cheongsam is more suitable in the chest, waist, abdomen and buttocks, and the overall effect of the garment is satisfactory.

In this paper, the fuzzy set theory is applied to the cheongsam model modification, and the modified value of each part of the cheongsam is obtained by using the fuzzy membership function formula, which improves the fit of the cheongsam. Therefore modified cheongsam pattern is more suitable for the garment production. Some scholars used a parametric garment model to perform a 3D simulation of the men's jacket of the closest garment size on the selected body model, and analyzed the model's looseness modification based on the

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computer's cross section for suitability evaluation (Mahnic and Petrak, 2013). This method only considers the modification of the cross section, and does not take into account the linkage relationship between the various parts of the garment and the consistency between the curves. When performing the modification of the pattern parameters, it uses the 3D simulation virtual verification. If the pattern can be outputted as a garment for real verification, its credibility will be greatly improved.

*Virtual fitting test result*

Through the virtual fitting software, the virtual model of the same size as the subject is dressed on the pattern before and after the modification to determine the fit of the cheongsam. Since the fabric used in the real production is a micro-elastic material blended with silk and spandex, the performance setting of the fabric in the software needs to be consistent with the real situation, so the maximum distance between the fabric and the human body is 4 cm. The display after the fabric distance setting is completed as shown in Plate 6.

After the performance setting of the fabric has been completed, the subject performs a virtual try-on on the 1#. Plate 7a is the virtual fitting result of the 6-yard cheongsam model, and Plate 7b shows the distance between the garment and the human body. The color is from red yellow to green and finally to the blue, indicating that the distance between the garment and the surface of the human body is getting larger and larger.

From Plate 7a, the subject in a cheongsam in the front has a certain curvilinear beauty. The shape of the cheongsam on the side is obviously inconsistent, and the whole shape is close to the *H*-shape. The front of the subject in Plate 5b shows that the chest is surrounded by a significant fold in the abdominal circumference, and its color is basically red, indicating that the distance between the human body and the clothing is insufficient to meet the amount of

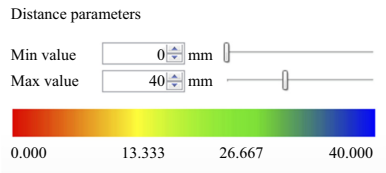


**Plate 6.**
Fabric distance setting

**(a)** **(b)**

**Plate 7.**
Cheongsam 6-code pattern virtual fitting

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cheongsam activity. The hips are blue and green, indicating that the distance between the human body and the clothing is large and needs to be improved. On the side of cheongsam, there is almost no gap between the chest and waist. On the back side, there are obvious folds on the back, indicating that the human body is too close to the clothing. It is necessary to increase the distance between the human body and the clothing. The hips are red and wrinkled. It is also necessary to increase the distance between clothing and the human body.

The modified cheongsam pattern is subjected to virtual fitting, and the result is shown in Plate 8. From Plate 8a, the modified cheongsam front body has a distinct curve; the back and waist and hip at the side shape show a certain curve of the human body, and there is no wrinkle at the chest, which is relatively smooth. It is obvious from Plate 8b that the body is basically yellow and green in the waist and abdomen, which not only satisfies the activity of the human body, but also fulfills the requirements of the cheongsam fit.

After setting the attribute of the fabric, the subject will perform a virtual try on the 2#. Through the virtual fitting software, the virtual model of the same size as the subject is dressed on the cheongsam pattern before and after the modification to determine the fit of the cheongsam initially. Plate 9a is the virtual fitting result of the 8 yard cheongsam pattern, and Plate 9b is the result of the distance between the garment and the human body surface. The

(a) (b)

**Plate 8.**
Cheongsam modified pattern virtual fitting

(a) (b)

**Plate 9.**
Cheongsam 8 yards pattern virtual fitting

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color is from red to yellow to green and finally to blue, indicating that the distance between the garment and the human body surface is getting bigger and bigger.

Plate 9 indicates that the cheongsam is obviously inconsistent, the whole shape is close to the *H*-shape, the circumference around the chest is basically yellow, and the chest-to-neck circumference is basically yellow-green, and the distance between the human body and the clothing is obviously greater than 2 cm. From the side of the cheongsam, there is almost no gap between the chest and waist, and the area between the back wide line and the hip line is basically green and blue. The distance between the human body and the clothing is obviously greater than 4 cm.

The modified cheongsam pattern is used for virtual fitting, and the result is shown in Plate 9. The garment is closely attached to the human body in the chest, waist and buttocks in Plate 10a. Except for the necessary looseness of the body part, the rest are almost red areas in Plate 10b, indicating that the amount of clearance between the modified garment and the human body is small, and the cheongsam is relatively close-fitting.

In this paper, the fit of the virtual try-on evaluation cheongsam is determined according to the distance between the human body and the cheongsam, and the difference between the human body and each part of the garment can be directly obtained. Slavenka Petrak used a 3D simulation of the closest garment size on the selected body model, verifying that each subject's adjustment mode was a fitting on a visual 3D body model. The values of the circumference of the body and the clothes on the cross section are determined, and the amount of relaxation is calculated based on the difference in the obtained values (Petrak *et al.*, 2015). The amount of relaxation is calculated by calculation, and it is more convenient if the data is directly observed.

## Conclusions and future work

In order to realize the industrialization of cheongsam, this paper proposes a method to calculate the amount of cheongsam pattern modification using fuzzy set theory. Through the analysis of the grading rules and the characteristics of the cheongsam pattern, the modified linkage parts and control points of the cheongsam pattern have been defined. Based on fuzzy set theory and human body shape analysis, the cheongsam pattern modification membership function was established. In this paper, taking the two experimental subjects an examples, the closest size is found in the cheongsam customer size table, and the difference between the subject and the size is obtained. The modified value of each joint part of the cheongsam is calculated by the membership function, and a customized pattern of the cheongsam fit is

(a) (b)

**Plate 10.** Cheongsam modified pattern virtual fitting

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obtained. At last, through the comparison of 3D virtual try-on and real-life fitting, the dressing state of the cheongsam before and after the modification of the model was experimented, and the effectiveness and conformity improvement quality of the cheongsam model modification method were verified.

The cheongsam pattern modification method given in this paper can meet the requirements of rapid prototyping in industrial customization, but the pattern modification method of membership function has a strong correlation with the model structure, and the membership function corresponding to various human body types is also different. This experimental method is more applicable to companies which have sample accumulation, because they already have corresponding basic patterns for various fabrics or clothing styles. In the future, the clothing of different fits will be considered and experimented, and the modification method of customized clothing pattern will be further improved.

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